

Experiment 1:

Introduction to Maven and Gradle: Overview of Build Automation Tools, Key Differences Between Maven and Gradle, Installation and Setup

Overview of Build Automation Tools

Build automation tools are software utilities used to **automate repetitive tasks** involved in software development such as:

- Compiling source code
- Managing project dependencies
- Running unit tests
- Packaging applications
- Deploying builds

In Java-based projects, build automation tools help developers maintain **consistency, faster builds, and easier project management**.

The two most popular build automation tools are **Apache Maven** and **Gradle**.

Apache Maven

Apache Maven is a **project management and build automation tool** primarily used for Java applications.

It uses an XML-based configuration file called **pom.xml** (Project Object Model).

Features of Maven:

- Standard project structure
- Dependency management using Maven Central Repository
- XML-based configuration
- Easy integration with IDEs and CI/CD tools

Gradle

Gradle is a **modern build automation tool** that combines the best features of Maven and Ant.

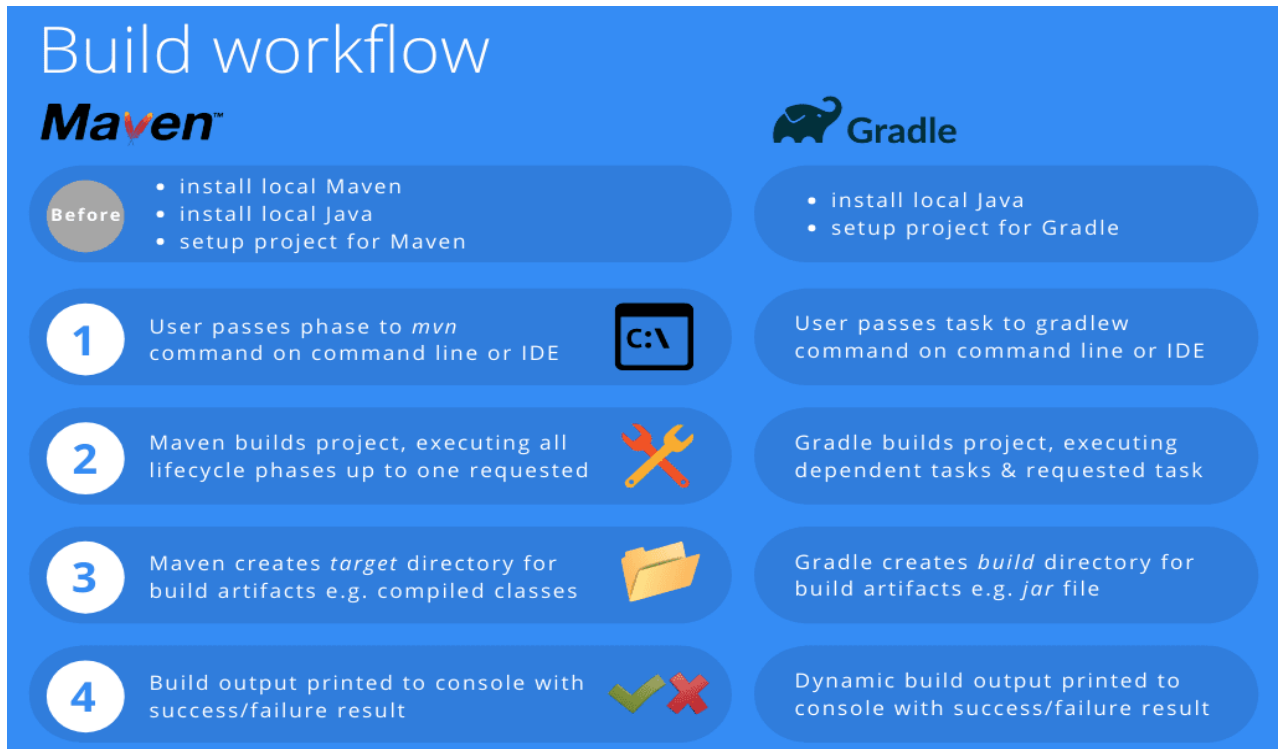
It uses **Groovy or Kotlin-based DSL** scripts for configuration.

Features of Gradle:

- Faster build performance using incremental builds

- Flexible and customizable build scripts
- Supports multi-project builds
- Strong dependency management

Key Differences Between Maven and Gradle



Installation and Setup

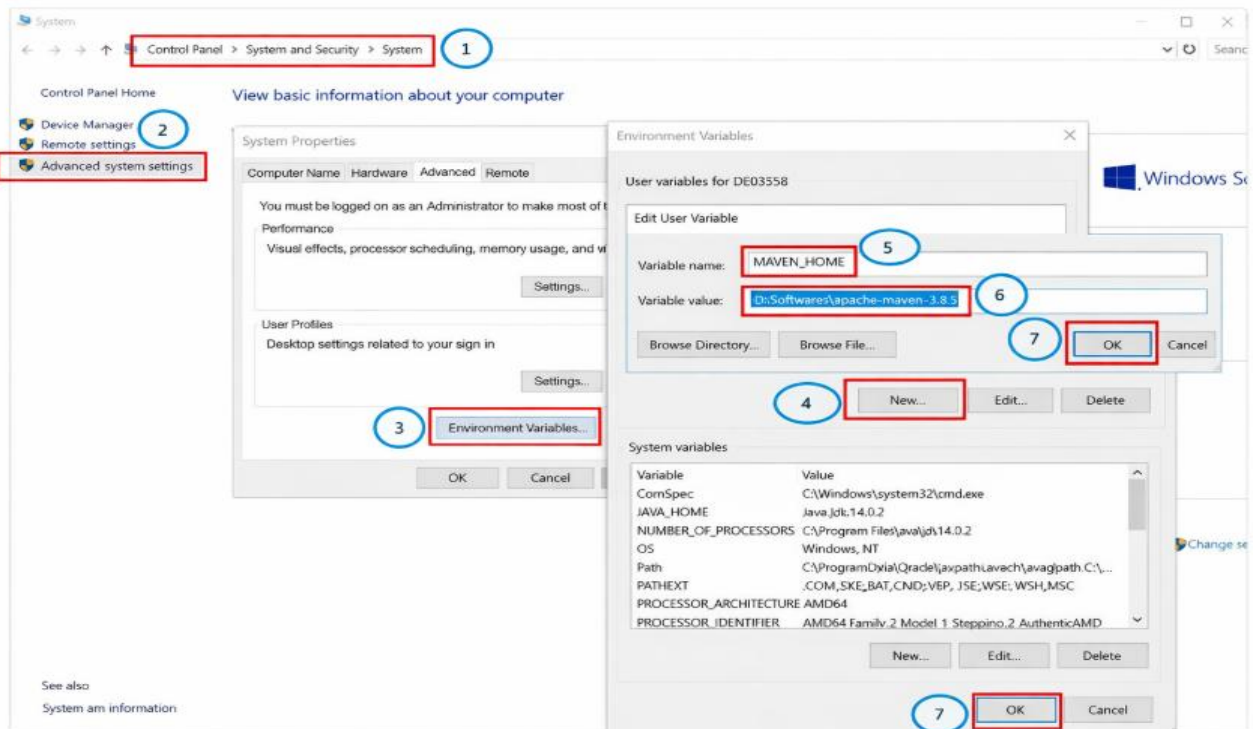
1. Installing Maven

Steps:

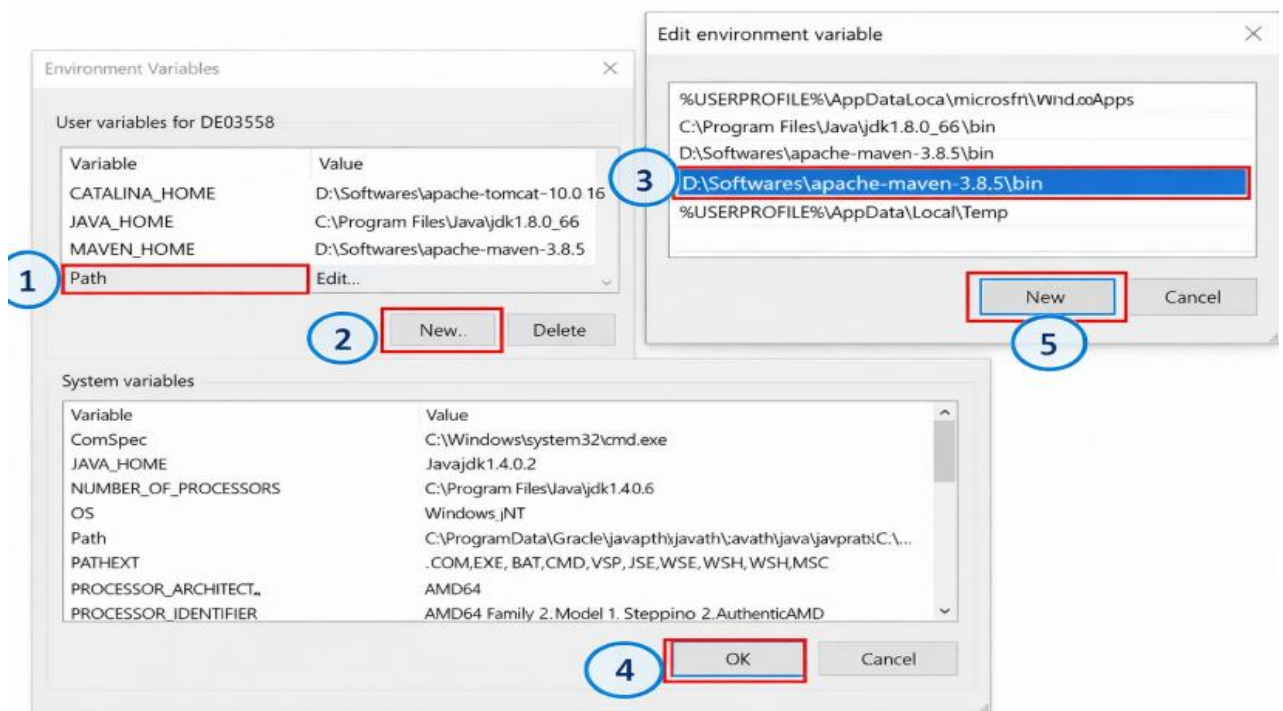
1. Download Apache Maven from the official website.

Link	Checksums	Signature
Binary tar.gz archive	apache-maven-3.8	apache-maven-3.8.5-bin.tar.gz
Binary zip archive	apache-maven-3.8.5-bin	apache-maven-3.8.5-bin.zip.sha512
Source tar.gz archive	apache-maven-3.8.5-src	apache-maven-3.8.5-src.tar.gz
Source zip archive	apache-maven-3.8.5-src	apache-maven-3.8.5-src.zip.sha512
Source zip archive	apache-maven-3.8.5-src	apache-maven-3.8.5-src.zip.sha512

2. Extract the archive to a directory (e.g., C:\Program Files\Apache\maven).
3. Set environment variables:
 - MAVEN_HOME



- Add Maven bin folder to PATH.



4. Verify installation:

`mvn -version`

If Maven has been setup correctly then we will get following output of the above command:

```
C:\Users\>mvn -version
Apache Maven 3.6.3 (cecedd343002696d0abb50b32b541b8a6ba2883f)
Maven home: D:\Softwares\apache-maven-3.6.3\bin\..
Java version: 1.8.0_66, vendor: Oracle Corporation, runtime: C:\Program Files\Java\jdk1.8.0_66\jre
Default locale: en_AU, platform encoding: Cp1252
OS name: "windows nt (unknown)", version: "10.0", arch: "amd64", family: "windows"
```

2. Installing Gradle

Steps:

1. Download Gradle from the official website.
2. Extract the files to a directory.
3. Set environment variables:
 - o GRADLE_HOME
 - o Add Gradle bin folder to PATH.
4. Verify installation:

gradle -version

1 Download Gradle from the official website



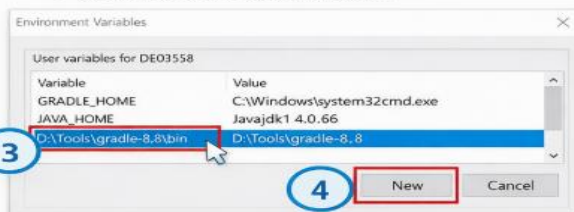
2 Extract the files to a directory

- GRADLE_HOME
- Add Gradle bin folder to PATH



3 Set environment variables:

- GRADLE_HOME
- Add Gradle bin folder to PATH



4 Verify installation:

- gradle -version

